

# IGZO eDRAM Atom Processor Footprint

https://gemini.google.com/app/d0bfc1d6bb7e18c9

User prompt: how much wafer space in mm would 128MB of IGZO BEOL eDRAM occupy with a single core Atom X6200FE

Response: To estimate the physical wafer area for this combination, we can break the footprint down into two main components: the logic core (**Intel Atom x6200FE**) and the monolithic **128 MB IGZO BEOL eDRAM** array.

## 1. Atom x6200FE Core Footprint

The Atom x6200FE is based on the **Tremont microarchitecture** (Elkhart Lake family), fabricated on Intel’s 10 nm process.

- **Full Dual-Core Die (Elkhart Lake x6200FE):** The full dual-core compute and I/O die is approximately 107 mm<sup>2</sup>.
- **Single Tremont Core Module:** A single Tremont core itself (excluding shared L2 cache, memory controllers, PCIe/I/O, and GPU logic) occupies roughly 1.2 mm<sup>2</sup> to 1.5 mm<sup>2</sup>.
- **Core + Local Cache / System Fabric Overhead:** Adding a dedicated 1.5 MB L2 cache and basic system interconnect brings the logic footprint for a isolated single-core domain to approximately 4 mm<sup>2</sup> – 6 mm<sup>2</sup>.

## 2. 128 MB IGZO BEOL eDRAM Area

Because IGZO (Indium Gallium Zinc Oxide) transistors are integrated into the **Back-End-of-Line (BEOL)** metallization layers over the silicon substrate, they offer distinct area advantages:

- **2T0C (2-Transistor, 0-Capacitor) Architecture:** Typically achieves a density of roughly 0.015 μm<sup>2</sup> – 0.025 μm<sup>2</sup> per bit depending on the process node (e.g., scaled 28nm or 22nm BEOL process).
- **1T1C / Scaled BEOL DRAM Architecture:** Yields a bit-cell area of around 0.008 μm<sup>2</sup> – 0.012 μm<sup>2</sup>.

Calculating for **128 MB** (1,024 Megabits / 1.024 billion bits):

Bit Count = 128 × 1024 × 1024 × 8 = 1,073,741,824 bits

- **Raw Memory Array Area:**

Area = 1.073 × 10<sup>9</sup> bits × 0.015 μm<sup>2</sup>/bit ≈ 16.1 mm<sup>2</sup>

- **Peripheral Overhead (Sensing, decoders, routing ~25–30%):** Adds ≈ 4 to 5 mm<sup>2</sup>.
- **Total 128 MB eDRAM Footprint:** Roughly 20 mm<sup>2</sup> to 25 mm<sup>2</sup>.

## 3. Total Combined Wafer Footprint Summary

Because BEOL IGZO eDRAM builds vertically on top of standard interconnect metal layers, it can partially overlay non-critical FEOL logic (or sit adjacent to it).

| Component                                  | Estimated Area (mm <sup>2</sup> )         |
|--------------------------------------------|-------------------------------------------|
| Single Atom Tremont Core (Core + L2 Cache) | 4.0 – 6.0 mm <sup>2</sup>                 |
| 128 MB IGZO BEOL eDRAM Array + Periphery   | 20.0 – 25.0 mm <sup>2</sup>               |
| Un-overlapped Silicon Footprint Total      | ≈ 24 mm <sup>2</sup> – 31 mm <sup>2</sup> |

Note: If the IGZO memory array is integrated directly over the underlying Atom core logic layers, the net increment on total silicon substrate area is strictly dominated by the size of the 128 MB memory array footprint (~ 20 – 25 mm<sup>2</sup>).

### Sources

1. [Intel Atom® x6200FE Processor](#)

User prompt: is this with 2D Planar 2T0C?

Response: Yes, that estimate assumes a standard single-layer 2D Planar 2T0C bit-cell design.

If you adjust the architectural configuration of the IGZO BEOL array—such as using 3D vertical stacking or switching to a 1T1C structure—the array footprint shifts as follows:

- **2D Planar 2T0C (20 mm<sup>2</sup> – 25 mm<sup>2</sup> total array):** Assumes both thin-film transistors (Write and Read) lie flat side-by-side on a single BEOL metallization plane. Cell sizes typically range from  $2 \times (2F \times 4F) \approx 16F^2$  to  $24F^2$ . At modern sub-30nm BEOL pitch, this yields a bit-cell area around 0.015 μm<sup>2</sup> – 0.020 μm<sup>2</sup>.
- **3D Vertically Stacked 2T0C (10 mm<sup>2</sup> – 14 mm<sup>2</sup> total array):** If the write transistor is stacked vertically directly on top of the read transistor within the metal layers (a 3D vertical IGZO channel layout), the effective cell footprint shrinks by roughly 40%–50% down to  $\approx 8F^2 - 10F^2$  (0.007 μm<sup>2</sup> – 0.010 μm<sup>2</sup> per bit).
- **Vertical 3D-DRAM / Multi-layer Stacking (< 8 mm<sup>2</sup> total array):** Because IGZO can be deposited at low BEOL processing temperatures (< 350–400 °C) without damaging underlying FEOL silicon, you can stack multiple 2T0C layers on top of each other in the back-end. A 2-layer or 4-layer stacked IGZO array drops the physical wafer area occupied by the 128MB cache proportionally down to  $\sim 6 \text{ mm}^2 - 10 \text{ mm}^2$ .

Revised Total Wafer Comparison

| IGZO BEOL 2T0C Layout Type      | Estimated 128MB Memory Area | Total Chip Footprint (With Atom x6200FE Core) |
|---------------------------------|-----------------------------|-----------------------------------------------|
| 2D Planar Single-Layer          | 20 – 25 mm <sup>2</sup>     | ≈ 24 – 31 mm <sup>2</sup>                     |
| 3D Vertically-Folded Transistor | 10 – 14 mm <sup>2</sup>     | ≈ 14 – 20 mm <sup>2</sup>                     |
| 2-Layer Multi-Story Stacked     | 10 – 12 mm <sup>2</sup>     | ≈ 14 – 18 mm <sup>2</sup>                     |

(Note: If the IGZO array is placed directly **above** the Atom core in the BEOL stack rather than adjacent to it on the silicon substrate, the core footprint and memory footprint share the same surface area, so only the larger of the two determines the total silicon die size.)

Sources

1. [Ultra-low Leakage IGZO-TFTs with Raised Source/Drain for V t > 0 V](#)

User prompt: yes

Response: Integrating a 128 MB IGZO BEOL eDRAM directly over the Atom x6200FE core introduces significant trade-offs between thermal dissipation, routing density, and performance.

1. Thermal Impact on the Atom Core

Standard silicon dies dissipate heat outward through the package lid (IHS). Inserting an IGZO eDRAM layer into the Back-End-of-Line (BEOL) metal stack changes the thermal profile in three main ways:

- **Thermal Blanket Effect:** Amorphous IGZO has a low thermal conductivity ( $\approx 1.5 - 2.5 \text{ W/m} \cdot \text{K}$ ) compared to bulk silicon ( $\approx 145 \text{ W/m} \cdot \text{K}$ ). Surrounding inter-metal dielectrics (low-*k* dielectrics) act as thermal insulators. Stacking memory over the 10 nm Tremont core traps heat in the front-end (FEOL) logic, causing local hotspots to run higher in temperature.
- **Leakage and Throttling:** Operating at higher steady-state temperatures increases core static power leakage. To prevent thermal runaway, max clock speeds may require down-throttling by roughly **5% to 15%** under sustained multi-threaded or AVX workloads.
- **Low Memory Self-Heating:** A major benefit of IGZO eDRAM is its ultra-low off-state leakage ( $< 10^{-18} \text{ A}/\mu\text{m}$ ) compared to standard Si-SRAM. As a result, the eDRAM array itself generates minimal active self-heating, making the core beneath it the primary source of thermal stress.

2. BEOL Routing Complexity & Signal Integrity

Placing 1.07 billion IGZO bit-cells above the logic core consumes metal layers typically used for global interconnects, power delivery, and clock trees:

- **Metal Layer Allocation:** Standard advanced processes utilize 10 to 14 metal layers (M1–M14). IGZO TFTs and their wordline/bitline arrays occupy intermediate layers (e.g., M4–M7). This forces core power delivery networks (PDN) and global clock signals to route around or through the memory array.
- **Through-Memory Vias (TMVs):** Power and ground must reach the underlying Atom core through dense power vias passing directly through the IGZO memory block. This reduces available routing tracks for local eDRAM interconnects and increases *IR*-drop across the core logic.

- **Parasitic Capacitance:** Signal lines running immediately adjacent to the high-density eDRAM array experience increased line-to-line parasitic capacitance, which can impact signal slew rates and increase dynamic power consumption along high-speed buses.

Summary Architectural Trade-off

| Design Metric         | Direct Vertical Overlay (M3D Stacking)                     | Adjacent Side-by-Side Placement                    |
|-----------------------|------------------------------------------------------------|----------------------------------------------------|
| Total Die Footprint   | $\approx 20 - 25 \text{ mm}^2$ ( <i>Memory-dominated</i> ) | $\approx 24 - 31 \text{ mm}^2$ ( <i>Combined</i> ) |
| Core Thermal Overhead | High (Heat trapped under low- $k$ /IGZO layers)            | Low (Direct path to heat sink)                     |
| Routing Complexity    | High (Requires TMVs, congested mid-metals)                 | Low (Standard BEOL metal routing)                  |
| Latency / Bandwidth   | Ultra-Low Latency (Short vertical vias)                    | Slightly Higher (Longer lateral buses)             |

Sources

1. [Thermal analysis of advanced back-end-of-line structures ... - Lirias](#)

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